Higher National Diploma in Engineering (Electrical)
Fourth Year, First Semester Examination – 2020
EE 3212 – Broadcast Technologies

Instructions for Candidates:
Answer any **FIVE (05)** questions
All questions carry equal marks

No. of questions: **06**
No. of pages : **03**
Time : **(03) Three hours**

01.

i. Explain 'Stereophonic sound' in brief. **(02 Marks)**

ii. What are the advantages and disadvantages of Stereo Recording? **(03 Marks)**

iii. What is the importance of public broadcasting? **(04 Marks)**

iv. Explain 'ZENITH-GE' system for FM Broadcasting and draw the frequency spectrum of the composite FM Stereo signal. **(05 Marks)**

v. Give the following specifications for FM broadcasting in Sri Lanka. **(06 Marks)**

Parameter	Figures and Details
Frequency Band	
Channel Width	
Max Deviation	
Audio Subcarrier Frequency	
Stereo Centre Frequency	
RDS Carrier Frequency	

[Total 20 Marks]

02.

i. Describe 'Pilot Tone'. **(02 Marks)**

ii. Draw the block diagram of VHF FM stereo encoder and briefly how it works mentioning the functions of following blocks,

a. Pre-emphasis
b. Matrix
c. Frequency Doubler

(06 Marks)

iii. Define positive and negative modulation. **(04 Marks)**

iv. Draw the block diagram of a stereo FM receiver indicating three major sections (Mono mode section, Stereophonic mode section and Common section for both Mono and Stereo modes) separately. **(05 Marks)**

v. Briefly explain 'Cross talk' between L and R channels. **(03 Marks)**

[Total 20 Marks]

03.

i. Briefly describe hue and saturation. **(03 marks)**

ii. What is photoconductivity? **(03 marks)**

iii. What is known as flicker? **(03 marks)**

iv. What is the main function of the blanking pulses? **(02 Marks)**

v. Define picture resolution. **(02 Marks)**

vi. Among colour difference signals, (G-Y) is not chosen for transmission. Give reasons for that. **(02 Marks)**

vii. Draw a simplified block diagram of monochrome TV camera tube and briefly explain its operation. **(05 Marks)**

[Total 20 Marks]

04.

i. Explain how a monochrome TV transmitter works with a suitable diagram. **(05 Marks)**

ii. Explain why separate colour sync pulses (colour burst) are needed in the colour TV System. **(03 Marks)**

iii. Why back porch is longer than front porch? **(02 Marks)**

iv. What is 'Aspect Ratio' and what are the reasons for selecting it as 4/3? **(04 Marks)**

v. In monochrome television what are the three main components in Composite Video Signal (CVS) and State the purpose of these three signal components. **(03 Marks)**

vi. Why the neck of the picture tube is made narrow? **(03 Marks)**

[Total 20 Marks]

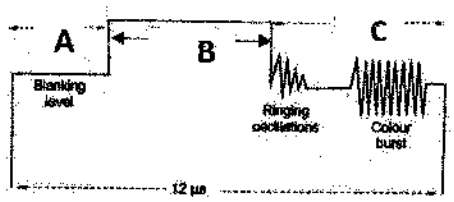
05.

i. Why is scanning necessary in television system? **(04 Marks)**

ii. Explain *'Interlaced scanning'*. **(03 Marks)**

iii. Why is the number of scanning lines in a frame always odd? **(04 Marks)**

iv. The following figure shows the 'Horizontal Blanking Pulse'. Name A, B and C with their relevant time periods. **(03 Marks)**



v. Write short notes on following terms. **(02x3 = 06 Marks)**

(a). Luminance

(b). Shadow mask

[Total 20 Marks]

06.

i. What are the applications of Digital Signal Processing (DSP)? **(03 Marks)**

ii. What are the advantages and disadvantages of DSP systems? **(04 Marks)**

iii. Compare Finite-duration Impulse Response (FIR) filters with Infinite-duration Impulse Response (IIR) filters. **(04 Marks)**

iv. What are the steps required to design an IIR digital filters? **(03 Marks)**

v. Write short notes on following terms. **(02 x 3 = 06 Marks)**

a) Pulse Code Modulation (PCM)

b) Differential Pulse Code Modulation (DPCM)

c) ADPCM: Adaptive Differential Pulse Code Modulation

[Total 20 Marks]